

PackSmart: Object Dimension and Weight Estimation From Multi-View 2D Images for Automated Packaging Box Design

Vedanti Pacharne
Dept. of Computer Engineering
PVGCOET&M, Pune
Pune, India
vedantipacharne@gmail.com

Sayee Rananaware
Dept. of Computer Engineering
PVGCOET&M, Pune
Pune, India
sayee.rananaware@gmail.com

Aabha Thoke
Dept. of Computer Engineering
PVGCOET&M, Pune
Pune, India
aabhathoke6@gmail.com

Manaswi Dasi
Dept. of Computer Engineering
PVGCOET&M, Pune
Pune, India
dasi.manaswi@gmail.com

Prof. A. G. Dongre
Dept. of Computer Engineering
PVGCOET&M, Pune
Pune, India
agd_comp@pvgcoet.ac.in

Abstract—Traditional methods for estimating volumes focus primarily on 3D scanning and measurement technologies. These methods include end devices that may not be suited for real-time volumetric measurement in an economically efficient manner. This paper proposes a system called PackSmart that provides packaging recommendations by estimating the geometry and material properties of objects based on 2D multi-view images captured by the user. The recommended system incorporates a Deep Learning based Object Detection model (YOLOv12) [1], to capture the coordinates of bounding boxes on captured images of the object in the front and top views. The model uses pixel-to-metric ratio conversion to estimate real-world levels based on size scaling which is determined empirically. The volume is calculated by these obtained dimensions. The system incorporates an EfficientNet-B4 based Convolutional Neural Network (CNN) [2] to classify material types, which is then used for a density-based weight estimation by the system. The combination of these properties of the object leads to the application of optimization rules by the system to recommend dimension and materials of boxes with protective layers. In addition to these recommendations, the system also provides a "Materials List" You can get the results and also download the "Pricing Documents" for more information of the experimentation prove that the method that has been proposed does provide the ability to estimate dimension and material properties of an object in an accurate manner without the need to utilize costly 3D measurement systems.

Index Terms—Industrial Packaging, Computer Vision, YOLOv12, EfficientNet-B4, Object detection, Volume and Weight estimation, Material classification

I. INTRODUCTION

The e-commerce and logistics business is growing rapidly in the past few years, and hence there is a need to automate the process of efficiently packaging a variety of goods. In

majority of warehouses globally, packaging decisions are made manually or through semi-automated methods.

Studies show that poor size estimation alone can cause 20–30% excess material usage in conventional packaging workflows. [3] When considered at scale, thousands of packages are processed daily. Therefore, small efficiencies add up to cause greater problems.

While object detection and material classification techniques have individually shown good results in research, they are rarely combined into a single, packaging-focused pipeline. Most industrial solutions rely on 3D scanners and depth sensors to get the product properties. The equipments are expensive and not suitable for small businesses. [4]

Therefore, this paper introduces PackSmart - a vision-based system that works with ordinary 2D images taken from two angles. It estimates an object's dimensions, material type, and weight without needing any specialized hardware.

II. LITERATURE SURVEY

Building PackSmart required drawing from multiple research areas. This section summarizes relevant prior work in object detection, dimension estimation from images, weight prediction, material classification, and packaging optimization - and explains how each area influenced our design choices.

A. Object Detection

Faster R-CNN by Ren et al. [5] was able to detect objects accurately, but the two-step process made it too slow for anything that needed to run in real time.

YOLO [6] flipped that around. It looks at the whole image at once and predicts boxes and class labels in a single pass. Much faster. YOLOv3 [7] added the ability to detect objects at different scales. We went with YOLOv12 [1] for this

project - it uses an attention mechanism that squeezes out better accuracy while keeping inference speed fast enough for practical use.

B. Dimension and Volume Estimation from 2D Images

The core problem with 2D images is that a photo is flat - you lose depth the moment you press the shutter. The old workaround is putting a ruler or known object in the frame and using it as a reference scale [10]. Lee et al. [11] modernized this by pairing it with the Segment Anything Model, which lets you get accurate measurements without any depth sensor at all.

Our approach is similar but simpler. The user just tells us one real measurement - the object's actual width. We use that to calculate a scale factor and apply it to both views. It works surprisingly well without any external object or hardware.

C. Weight Estimation

Weight is the hardest part to estimate because you cannot see it. Standley et al. [13] showed that a CNN can actually learn to guess mass from visual cues like texture and shape - the network picks up on things humans would not consciously notice. Konovalov et al. [14] used a two-photo approach for estimating fish weight, combining geometry with known density values for each species. Wimalasiri et al. [15] did something similar for food.

We borrowed that density-based logic. The detected material's density is to be multiplied with the estimated volume. For hollow things like clocks or bottles, we apply a fill factor so that the module is not wildly overestimating just because the bounding box is large.

D. Material and Texture Classification

Bell et al. [16] put together the MINC dataset specifically for material recognition and showed that fine-tuned CNNs are significantly better than older hand-crafted texture approaches on real-world images. Li et al. and Liu and Wang later showed that deep residual features hold up well across metals, ceramics, plastics, and organic materials.

We used EfficientNet-B4 [2] because it scales depth, width, and resolution in a balanced way, so you get strong performance without the bloat of a full ResNet [17] or VGGNet [18]. On our dataset of ten material classes it hit 86.2% validation accuracy, which was enough for reliable weight estimation.

E. Automated Packaging Optimization

Bortfeldt and Wäscher [19] did a thorough review of bin-packing algorithms and the main finding is that optimal packing for irregular shapes is genuinely hard, but in this system we were designing box so the shape of objects did not matter much. This approach helped us in applying protective layers. Heininger et al. [20] took a machine learning approach, training on historical packaging records to predict the right box size efficiently. DHL's OptiCarton is a real-world example showing that getting box sizes right actually moves the needle on cost and emissions.

We kept our packaging logic rule-based - weight and fragility determine the box type and padding. It is transparent, easy to adjust, and works well for the object categories we support.

III. METHODOLOGY

The proposed model estimates the dimensions, volume, material type, and weight of objects using two 2D images because a single image cannot provide all three spatial dimensions. Therefore, the system uses two orthogonal views (front and top). The images are to be captured at 90° viewing angle with respect to the camera to reduce distortion and improve measurement accuracy.

At present, the model is built for 33 objects commonly used in daily life, including electronic items, furniture, glassware, and various kitchenware products. The complete workflow is divided into multiple functional modules as follows:

A. Object Detection and Dimension Extraction

The PackSmart system performs object detection using two approaches based on the image perspective:

- Front View: YOLO-based deep learning detection.
- Top View: OpenCV contour-based detection,

The goal of this module was to detect an object and extract its coordinates, which will help calculate the dimensions. Initially, the module was designed to process a single image containing an isometric view of the object. By this view, all three dimensions can be extracted. But the existing depth models fail to estimate depth between two points in an image. This task becomes harder when the object is of irregular shape. Therefore, we proceeded by accepting two images from different angles.

A custom-trained YOLOv12 model was used for detection from the front-view image. The bounding box was drawn around the image to get the coordinates, which determined the height and length of the object. These dimensions were in pixel units. To train the model, the dataset is prepared with the set of objects that are frequently packaged. The dataset is prepared with 200 images for every object class. Before the training process, the images were resized to the input resolution of 640 * 640 pixels. The prepared dataset is divided into training (80%), testing (15%), and validation (5%) subsets.

The model achieved an mAP@0.5 score of 0.894. As shown in Fig. 1, the confusion matrix exhibits strong diagonal dominance. It indicates that most objects are correctly classified and not confused with other classes as per-class prediction accuracy lies predominantly between 90% and 100%. Analysis of the Precision-Recall curve indicates a favorable trade-off between precision and recall, while the F1-Confidence curve shows a maximum F1-score of 0.87 at a confidence threshold of 0.376, which was selected as the optimal operating threshold for deployment.

The custom-trained YOLO model fails to recognize the object from a top view. Because the dataset did not contain top views of objects. To extract length, the bounding box around the top-view image is needed. Therefore, the top

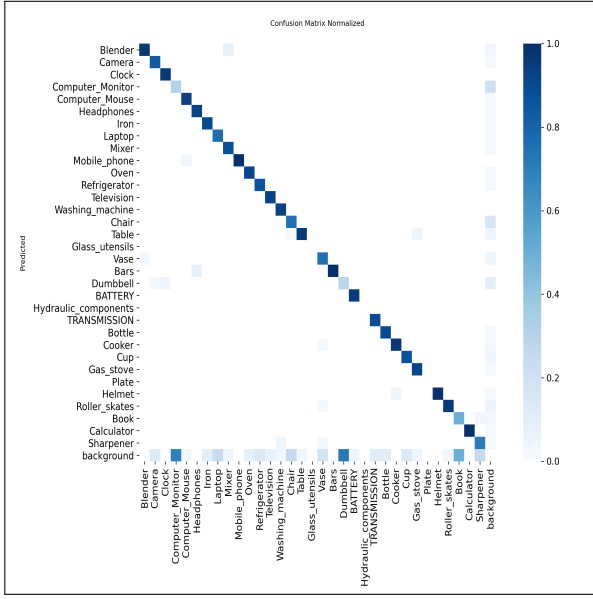


Fig. 1: Confusion Matrix

view image is processed using classical image processing techniques instead of deep learning. The preprocessing steps are applied sequentially:

- 1) Grayscale transformation is done on the RGB image.
- 2) A Gaussian blur filter of size 5x5 is used.
- 3) The threshold values of 50 and 150 are utilized for clever edge detection
- 4) Dilation with a 5x5 filter is performed to bridge small openings and strengthen edges around objects.

Then, External contours are extracted. In case of multiple contours being detected, the largest contour is selected based on the area. This is a reasonable assumption since only a single object is present in the image.

Bounding rectangle calculation is done for the largest contour:

$$(x, y, w, h) = \text{boundingRect}(C_{\max}) \quad (1)$$

where w and h are width and height measured in pixels.

They represent top-view dimension of the object.

Therefore, the object detection algorithm has successfully obtained the object dimensions measured in pixels. The pixel-based measurements will be used as inputs for scaling the dimensions.

B. Real-World Dimension Estimation

This module focuses on converting the pixel units into real-world dimensions (in centimeters). Since pixel values depend on camera resolution and distance, a scaling factor is required for accurate conversion. [12]. The width in cm is accepted from the user as a reference to set the scaling factor. Distance between camera and object being a factor in scaling of pixels to centimeters. Two scale factors are calculated for two images as they are captured from different perspectives. The scale factor is calculated as:

$$S_{front} = \frac{W_{real}}{W_{px}^{front}} \quad (2)$$

$$S_{top} = \frac{W_{real}}{W_{px}^{top}} \quad (3)$$

where W_{real} is width in cm, W_{px}^{top} is width in pixels obtained from the top-view image while W_{px}^{front} from the front view and S_{front} and S_{top} are scale factors.

Using this scale factors, the real-world length and height are computed as:

$$L_{real} = L_{px}^{top} \times S_{top} \quad (4)$$

$$H_{real} = H_{px}^{front} \times S_{front} \quad (5)$$

Thus, the module computes the real-world length, width, and height of the object which are further used to calculate the volume.

$$V = L_{real} \times W_{real} \times H_{real} \quad (6)$$

C. Material Estimation

For material estimation, we use a deep learning-based image classification system. An EfficientNet-B4 convolutional neural network is chosen due to its superior feature extraction capabilities. The cropped object image provided by front-view detection module will be fed into the material estimator as an input. Image will be preprocessed according to the requirements of the model in which:

- Change color format to RGB
- Resize to 224×224 pixels
- Convert into tensor format
- Normalized using ImageNet mean and standard deviation values

The normalization operation is expressed as:

$$I_{\text{norm}} = \frac{I - \mu}{\sigma} \quad (7)$$

where μ and σ represents channel-wise mean and standard deviation.

We modify the architecture of EfficientNet-B4 by replacing its last layer with an equivalent fully-connected layer having '10' number of classes.

We train our model for classifying following objects:

Aluminum', 'Brass', 'Copper', 'Iron', 'Steel', Ceramic, Glass, Paper, Plastic, Wood.

In the process of training, we found out that training and validation loss decreased consistently. While the validation accuracy increased progressively and reached a peak value of 86.20%. No further enhancement in validation accuracy was observed after reaching peak value. Training accuracy has reached about 89.9% while validation accuracy got stabilized in range of 85-86%.

Given an input image I, the neural network outputs the following output vector:

$$z = [z_1, z_2, \dots, z_{10}]$$



Fig. 2: Training and validation performance of EfficientNet-B4 model.

Softmax Activation is used to calculate the probability of each class:

$$P_i = \frac{e^{z_i}}{\sum_{j=1}^{10} e^{z_j}} \quad (8)$$

The material class M is calculated by using:

$$M = \arg \max(P_i) \quad (9)$$

The ultimate output from this block is the material label, which will be sent to the weight estimation block for calculating the weight of the object based on density.

D. Weight Estimation

With respect to the object volume, material type, and fill factor of the object, the weight estimation block calculates the object weight. A fill factor is required to take into consideration hollow objects since some objects have holes in them.

1) *Material Density Database*: A predefined density database is maintained for materials as shown in Table 1. The density values are stored in kg/cm^3 .

TABLE I: Material Density Database

Material	Density (kg/cm^3)
Aluminum	2.7
Brass	8.73
Copper	8.96
Iron	7.85
Steel	7.87
Ceramic	2.4
Glass	2.5
Paper	0.8
Plastic	1.2
Wood	0.7

2) *Object Fill Factor*: To improve estimation accuracy, an object-specific fill factor F is introduced which represents the proportion of the object's bounding volume that is occupied by material.

$$0 < F \leq 1$$

For example, objects such as mobile phone have a high fill factor, whereas bottles or refrigerators have lower fill factors due to hollow interiors.

3) *Effective Volume Calculation*: The effective material volume can be determined as follows:

$$V_{\text{effective}} = V \times F \quad (10)$$

where:

- V = calculated volume (cm^3)
- F = fill factor of the object

4) *Weight Computation*: The estimation of the object's weight can be made according to the following formula:

$$W = V_{\text{effective}} \times \rho_{\text{material}} \quad (11)$$

where:

- W = estimated weight (kg)
- ρ_{material} = material density (kg/cm^3)

Output of the module includes the estimated object weight in kilograms, rounded to three decimal places. Packing recommendations module utilizes the estimated object weight.

E. Packaging Logic

The system handles the packaging logic with a rule-based decision system that covers the most important real-world factors: how heavy the object is, and how fragile it is.

The first thing the system does is put the object into a weight category of light, medium, or heavy. Each category maps to a specific outer box material. Heavier objects need structurally stronger boxes, so the outer material changes accordingly. This part is straightforward and works well in practice.

Fragility is handled separately. The type of object and the level of fragility are checked independently of the weight of the object, and the type of material for the inner cushioning is decided accordingly. The fragile items are then assigned with bubble wraps, thermocol, or edge protectors depending on their fragility. Once the type of material for protection is decided, the system then calculates the final size of the box by adding padding on all sides of the box. The padding depends on the type of protection material allocated. This gives us the actual size of the box, which can hold the object along with the protection material.

Once the size of the box is calculated, the system generates two visual images. One is a 2D unfolded box layout, which is basically a flat cutout of the box showing all the parts of the box. The other is a 3D image of the assembled box, which allows for a better view of the design of the box before actually going into production.

Finally, the system puts together a Bill of Materials listing every component needed and generates a downloadable PDF quote. The whole thing can be downloaded, printed, or forwarded directly to a supplier.

IV. RESULTS

We checked our system on different objects such as a mobile phone, transparent glass vase, and a clock. All the objects were processed through the entire modules in the system. The results are as follows:

The estimated dimensions of the mobile phone are approximately 16.06 cm x 7.4 cm x 1.35 cm, while the actual dimensions are 16.1 cm x 7.4 cm x 1.3 cm. As a result, there is an average error in dimension calculation of approximately 3%. In terms of material classification, the material classification module recognized the object material as steel and glass. The program calculates that the weight would be 262 g, but the actual weight is 251 g, leading to a error rate of 4.3%. The weight by the system was calculated with the densities of steel and glass.

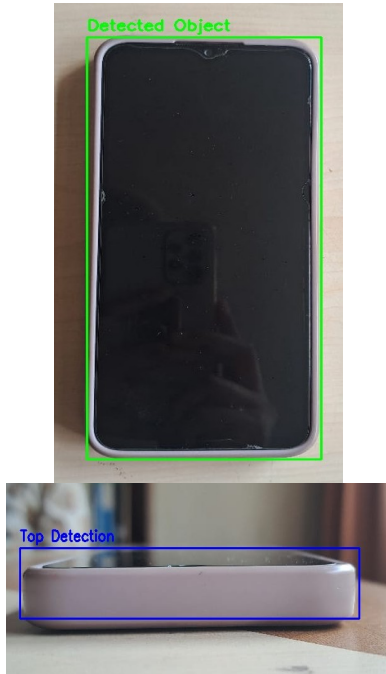


Fig. 3: Mobile Phone Detection

The dimension of the transparent glass vase has been calculated as 18.96 cm x 8.7 cm x 8.93 cm. The actual dimensions of the vase were 19.3 cm x 8.7 cm x 8.7 cm. In other words, there is an error of approximately 1.7-2.3%. For material classification, the material classification module recognizes glass material. From the calculated volume and the predefined density value of glass material, the program estimates the weight as 346 g while the actual weight is 332 g. The error rate in this case is 4%.

The system detected the clock and determined its size from both angles. The estimated size was around 20.52 cm x 20.5 cm x 4.41 cm. The real size was 20.5 cm x 20.5 cm x 4.3 cm, giving an error of about 2%. The material detector accurately classified the material as plastic, which is consistent with existing data. The estimated weight was 520 g, while the actual weight was 556 g, yielding an error of 6%.



Fig. 4: Vase Detection



Fig. 5: Clock Detection

The experimental results tell us that the proposed system can estimate dimensions with an error rate of 2-3%, and object weight with an error rate of 4-6%. These errors mainly arise from perspective variations in captured images, minor inaccuracies pixel-to-cm scaling, and the approximation introduced by fill factors for hollow objects. Despite these limitations, the system demonstrates reliable performance for automated packaging estimation and provides practical outputs for packaging design, including recommended materials, adjusted box dimensions, and packaging visualization.

V. CONCLUSIONS

This paper presented PackSmart, an automated system that automates packaging design by estimating object dimensions, material type, and weight from multi-view 2D images. The system combines all the modules like object detection, dimension extraction and weight estimation to generate suitable box designs according to the packaging rules. Based on the derived object properties, it recommends suitable packaging materials, produces 2D layouts and 3D visualizations, and outputs a Bill of Materials along with a downloadable quotation.

Overall, the system gives reliable solutions with minimal error rate which is suitable for implementing in industries for reducing human-prone errors in packaging process.

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